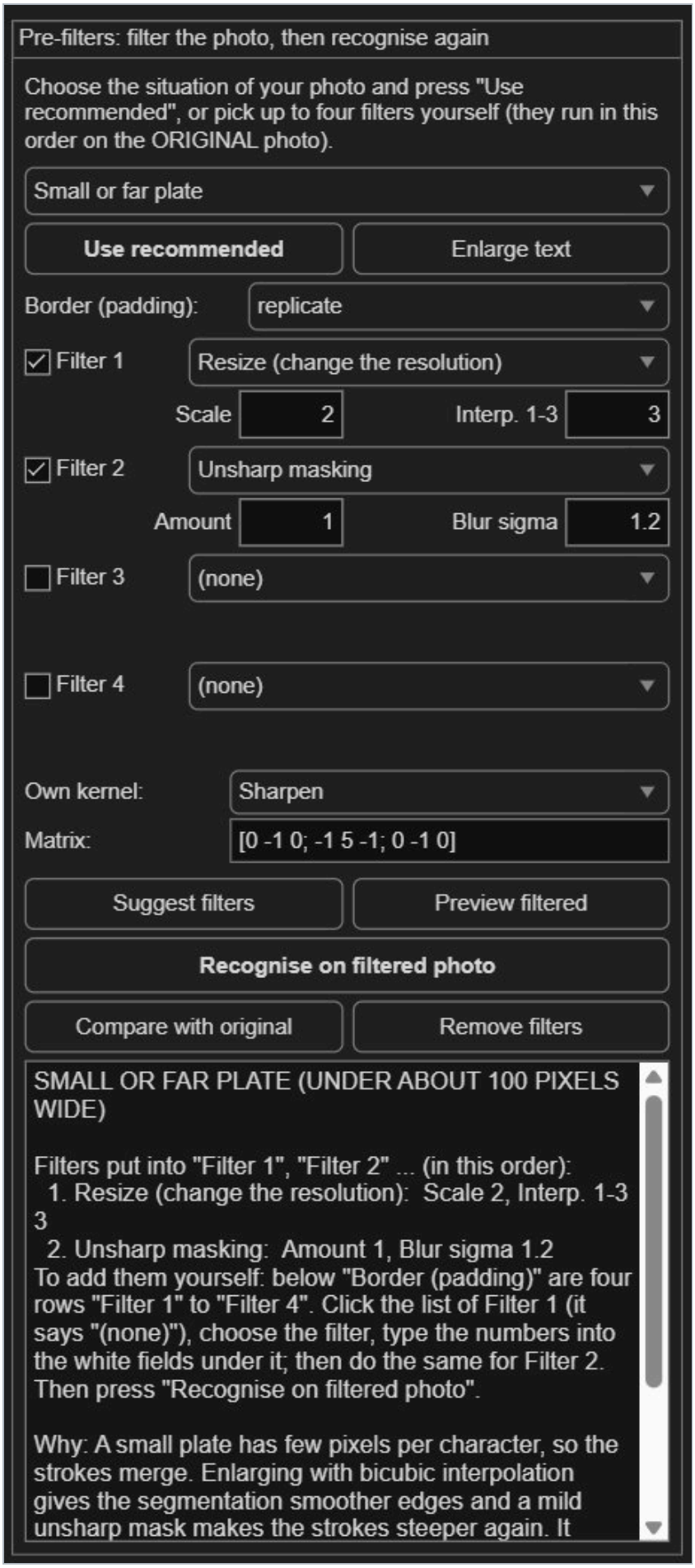
Filter guide for licence-plate recognition

Which filter for which photo: night, small and big plates, any perspective view. Measured with the MATLAB project "License Plate Recognition Using SVM" (October 2026).

A filter changes the pixels of the photo before the program looks for the plate. It helps only when the photo has the problem that the filter cures. The program already repairs brightness and contrast by itself, so most photos need no filter at all, and the wrong filter makes things worse: in the tests a Gaussian blur cost 3 of 16 plates, and on one of your own night photos an enlargement + sharpening turned a correct reading into a wrong one. For every situation this guide says which filters were tested, which helped and which did not.

How to use the filter panel

1. Start MATLAB, type clear all and then LPR_Toolbox (one line at a time). Press "Open image..." and choose your photo (JPG, PNG or JFIF).
2. Look at the photo and decide its situation: night? plate small or far (narrower than about 100 pixels)? big and close? seen from the side or from above? noisy, blurry, flat or shaded?
3. In the panel on the right open the list "Situation" and choose it. The text below the buttons tells you what to do. Press "Use recommended": the four filter rows "Filter 1" to "Filter 4" are filled in for you (for night, big and glare photos they are cleared, because no filter is the best choice). For photos with several cars it also ticks "Find ALL plates".
4. Press "Recognise on filtered photo". The plate is searched in the filtered photo. If nothing is found, the text says whether the original photo reads: then the filters made it worse.
5. Press "Compare with original" to see both readings. Keep the filter only if the reading gets better. "Remove filters" goes back to the original photo.
6. To add filters by hand: below "Border (padding)" there are four rows, "Filter 1" to "Filter 4". Click the list of Filter 1 (it says "(none)"), choose a filter and type the numbers into the white fields under it; the explanation of the filter appears in the text box. "Enlarge text" shows that text in a big window.



The filter panel on the right of the window.

Quick answer: which filter for which situation

Situation	Filters (in this order)	What was measured
Night view (dark photo, lamp glare)	**no filter**	No filter beat doing nothing (without a filter: lit street 8 of 16, very dark 6 of 16); the program now also searches an equalised copy by itself.
Small or far plate (under about 100 pixels wide)	1. Resize (change the resolution): Scale 2, Interp. 1-3 3 2. Unsharp masking: Amount 1, Blur sigma 1.2	4 -> 6 plates of 16 (photos shrunk to 40 %).
Big, close plate (over about 300 pixels wide)	**no filter**	No filter; shrinking the photo cost 3 plates (8 -> 5).
Angled plate (seen from the side)	1. Resize (change the resolution): Scale 1.5, Interp. 1-3 3 2. Unsharp masking: Amount 1, Blur sigma 1.2	Real oblique photos 1 -> 2 of 6; photos turned 30 degrees 5 -> 8, 50 degrees 4 -> 6 (of 16).
Seen from above (high camera)	1. Local histogram equalisation (CLAHE): Tiles 8, Clip limit 3	Simulated 30 degrees: local equalisation 5 -> 7 plates (fewer characters right); enlarging did harm (5 -> 4).
Night, small or far plate (street lit)	1. Resize (change the resolution): Scale 2, Interp. 1-3 3 2. Unsharp masking: Amount 1, Blur sigma 1.2	Lit street: 5 -> 7 plates of 16; very dark: 4 -> 2 (worse).
Many cars in the photo (traffic)	**no filter**	Ticks "Find ALL plates"; plates under 30 pixels are neither found nor read.
Many cars at night	**no filter**	Ticks "Find ALL plates"; your night traffic photo: 2 of 3 plates found.
Any vehicle: motorcycle, scooter, bus, truck, taxi	1. Resize (change the resolution): Scale 2, Interp. 1-3 3 2. Unsharp masking: Amount 1, Blur sigma 1.2	Ticks "Find ALL plates" and enlarges 2 times; two-line motorcycle plates: only one row.
White and black dots (salt-and-pepper noise)	1. Median filter: Window 3	5 -> 7 plates of 16 (but 5 points fewer characters right).
Grainy photo (Gaussian noise, high ISO)	1. Bilateral filter (edge-preserving): Spatial sigma 1.5, Range sigma 0.15	8 -> 9 plates; Gaussian blur 5, median 6.
Blurry photo (out of focus or shaky)	1. Unsharp masking: Amount 1.5, Blur sigma 1.5	5 -> 6 plates: small and not reliable.
Low contrast (haze, fog, rain, dirty glass)	1. Contrast stretching: Low % 1, High % 99	9 -> 9 plates; characters 81.2 -> 83.0 %.
Shadow or uneven light across the photo	1. Illumination correction: Blur sigma 40, Strength 1	6 -> 8 plates (characters 72.0 -> 70.3 %).
Glare or burnt-out plate (head-lamp, flash, sun)	**no filter**	Nothing to measure: take the photo again.
Photo of a screen (pixel grid, moire)	1. Frequency low-pass (noise): Cut-off 0.5	Not measured: open the original file instead.

Honest limits. All numbers come from only 16 held-out test photos: a difference of one or two plates is a hint, not a proof. The night photos in the tests are simulated (darkened, orange lamp colour, noise), not real

night photographs. No filter can create detail that was never recorded, bring back burnt-out white or read a plate that is hidden.

The situations, one by one

Night view (dark photo, lamp glare)

Filters to put into Filter 1, Filter 2 ...

```
No filter: the best choice is the original photo (leave Filter 1 to Filter 4 empty).
```

Why

The program repairs the brightness of a dark photo itself and also searches a locally equalised copy of it for the plate, so no filter is needed for the search. Filters added on top did not beat doing nothing on the test photos (they amplify the noise of a dark photo). What decides a night photo is the plate: about 100 pixels wide and not burnt out by a lamp. Plate smaller than that: choose "Night, small or far plate"; several cars: "Many cars at night".

What was measured

Measured on the 16 test photos made to look like night (darkened, orange lamp cast, two glows, noise), exact plates / characters. Lit street: none 7 / 62.3 %, bilateral 6 / 60.5 %, local equalisation 6 / 58.0 %, bilateral + local equalisation 6 / 60.6 %, gamma 0.6 6 / 66.5 %, median 7 / 62.6 %. Very dark: none 6 / 61.9 %, bilateral 5 / 53.8 % (4 / 48.9 % with a fixed sigma_r), local equalisation 5 / 46.4 %, bilateral + local equalisation 4 / 44.8 %, gamma 0.6 6 / 60.1 %, median 6 / 51.0 %. With the final code (night search + character splitting) doing nothing reads 8 and 6 plates (68.6 % and 61.9 % of the characters); the filters above were compared with the earlier code. These are simulated, not real night photographs.

Photo car_18 darkened like a lit street, orange lamp cast and noise (simulated). Red outline = plate accepted by the program.



none
reads LJ6J



bilateral
reads LJ6J



CLAHE
reads LJ6J



gamma 0.6
reads LJGJ

The same photo, very dark (simulated).



none
no plate



bilateral
no plate



CLAHE
reads LJG1



gamma 0.6
no plate

Small or far plate (under about 100 pixels wide)

Filters to put into Filter 1, Filter 2 ...

```
1. Resize (change the resolution): Scale 2, Interp. 1-3 3
2. Unsharp masking: Amount 1, Blur sigma 1.2
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

A small plate has few pixels per character, so the strokes of the characters merge. Enlarging with bicubic interpolation gives the segmentation smoother edges to work with, and a mild unsharp mask makes the strokes steeper again. It cannot add detail: below about 40 pixels of plate width nothing helps. Better: get closer, or use 'Pick car' on a bigger photo.

What was measured

Measured on the 16 test photos shrunk to 40 % (plates 58 to 240 pixels wide), exact plates / characters: none 4 / 58.0 %, enlarge x2 bicubic 5 / 59.4 %, enlarge x2 + unsharp 6 / 63.2 %, unsharp only 5 / 62.4 %. About +2 plates; with 16 photos this is a hint.

Photo car_12 shrunk to 40 % (the plate is about 90 pixels wide).



none
reads LAZL706



enlarge x2
✓ 4AZ4706



enlarge x2 + unsharp
✓ 4AZ4706

Photo car_09 shrunk to 40 % (the plate is about 114 pixels wide).



none
✓ 011CD228



enlarge x2
✓ 011CD228



enlarge x2 + unsharp
✓ 011CD228

Big, close plate (over about 300 pixels wide)

Filters to put into Filter 1, Filter 2 ...

```
No filter: the best choice is the original photo (leave Filter 1 to Filter 4 empty).
```

Why

A big plate already has plenty of pixels per character; filtering only risks blurring or adding noise. Use no filter. Do not shrink the photo to save time: that was the worst choice in the test.

What was measured

Measured on the 16 original test photos (1920 pixels wide), exact plates / characters: none 8 / 74.6 %, shrink to 0.5 5 / 61.4 %, Gaussian 3 x 3 7 / 73.7 %, unsharp masking 7 / 67.5 %.

Angled plate (seen from the side)

Filters to put into Filter 1, Filter 2 ...

```
1. Resize (change the resolution): Scale 1.5, Interp. 1-3 3
2. Unsharp masking: Amount 1, Blur sigma 1.2
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

An angled plate is mainly a shape problem: the characters are squeezed and slanted. The program straightens the plate itself (perspective correction), and no filter can do that; but enlarging 1.5 times gives the squeezed characters more pixels and a mild unsharp mask makes their strokes steeper. It helped on side views, but it is no cure for steep angles (from 50 degrees on most plates stay unreadable): photograph from the front if you can. Check with "Compare with original".

What was measured

Plates read exactly of 16 (characters correct), no filter / enlarge x1.5 + unsharp (1, 1.2): the 6 real oblique test photos 1 / 2 of 6; the 16 test photos turned 30 degrees to the side 5 (56.1 %) / 8 (65.0 %); turned 50 degrees 4 (48.2 %) / 6 (58.4 %). On the front-view photos the plates went from 8 to 9 but the characters from 74.6 % to 71.5 %. The turned photos are simulations (homography).

Seen from above (high camera)

Filters to put into Filter 1, Filter 2 ...

```
1. Local histogram equalisation (CLAHE): Tiles 8, Clip limit 3
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

From above the plate is squeezed vertically and the characters get short; the program straightens it itself. Local histogram equalisation gave more plates in the simulation but fewer correct characters, and enlarging + sharpening did harm here. Treat it as a try: press "Compare with original" and keep the filter only if the reading gets better.

What was measured

The 16 test photos turned 30 degrees about the horizontal axis (simulated). Plates read exactly of 16 (characters correct): no filter 5 (72.7 %); local equalisation 7 (66.0 %); enlarge x1.5 + unsharp 4 (59.8 %).

Night, small or far plate (street lit)

Filters to put into Filter 1, Filter 2 ...

```
1. Resize (change the resolution): Scale 2, Interp. 1-3 3
2. Unsharp masking: Amount 1, Blur sigma 1.2
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

A small plate at night needs the same help as a small plate by day: enlarging 2 times gives the characters more pixels and a mild unsharp mask makes their strokes steeper. The program also searches an equalised copy of a dark photo by itself. Use it only when the street is lit: for a very dark photo enlarging made things worse. Below about 40 pixels of plate width nothing helps; a plate lit red by the tail lights with lamp glare on its characters (your dark.jpg) is located but not read.

What was measured

The 16 test photos shrunk to 40 % and made to look like a lit street at night: no filter 5 plates (49.8 % of the characters), enlarge x2 + unsharp 7 (63.3 %), plus gamma 0.6 5 (56.0 %). Very dark photos: no filter 4 (43.6 %), enlarge x2 + unsharp 2 (28.5 %), plus gamma 0.6 2 (21.1 %): enlarging a very dark photo does harm. Check with "Compare with original".

Many cars in the photo (traffic)

Filters to put into Filter 1, Filter 2 ...

```
No filter: the best choice is the original photo (leave Filter 1 to Filter 4 empty).
```

Why

Ticks "Find ALL plates": every plate the program accepts is boxed (red: 90 pixels wide or more, orange: smaller, reading unreliable) and listed with its reading; plate-like regions that could not be read get grey dashed boxes. On a road photo with far cars the plates are 20 to 30 pixels wide: no filter makes them readable. A bigger photo or "Pick car..." (tab 4 Plate) is the cure.

What was measured

On the 16 test photos a region with a plate-SVM score of at least -0.3 contained the plate in 14 of 16 photos, also when shrunk to 40 %, with about 1 false region per photo. On your day traffic photo (640 x 480) the two nearest plates are only 23 and 26 pixels wide: nothing was found or read, with or without a filter (enlarging 2 and 3 times, gamma, local equalisation).

Many cars at night

Filters to put into Filter 1, Filter 2 ...

```
No filter: the best choice is the original photo (leave Filter 1 to Filter 4 empty).
```

Why

Like "Many cars in the photo", for a dark photo: every plate is searched, and an equalised copy of the dark photo is searched automatically. No filter is recommended (none helped on dark photos). Plates that are small, lit red or hidden in lamp glare are often found but read wrongly or not at all.

What was measured

One real night traffic photo of yours (1710 x 855, three plates): 2 plates found, the Mini T935AK797 (read right with every filter tried) and the dark SUV 6688 EP-2 (read 6688EP; 6688EP2 with local equalisation or enlarging 3 times, otherwise 6688F or 6688H); the white car (plate about 70 pixels) was missed. One photo is a hint only.

Any vehicle: motorcycle, scooter, bus, truck, taxi

Filters to put into Filter 1, Filter 2 ...

```
1. Resize (change the resolution): Scale 2, Interp. 1-3 3
2. Unsharp masking: Amount 1, Blur sigma 1.2
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

Ticks "Find ALL plates", because several vehicles are in the photo, and enlarges the photo 2 times: the small plate of a scooter or a far bus only shows up after that. The search is the same as for cars (trained on car plates). Two-line motorcycle plates are not supported: only one row is found and read. Buses and trucks with a car-like plate are treated like cars (not measured, no test photos). Also, a road sign with real letters can be read as a plate (a "NO TURN ON RED" sign gave TURN).

What was measured

Your Indian street photo: without a filter 1 plate read (the taxi, KA53C6332 for KA53C6932, one character wrong); with enlarge x2 + unsharp 2 (the taxi, KA53C63321, and the upper row of the two-line scooter plate KA02K read as K402X). One photo: a hint only.

White and black dots (salt-and-pepper noise)

Filters to put into Filter 1, Filter 2 ...

```
1. Median filter: Window 3
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

Isolated white or black dots cannot be the middle value of their neighbours, so a 3 x 3 median removes them and keeps the edges of the characters. Use 5 x 5 only when more than 8 % of the pixels are dots.

What was measured

Measured: 5 exact plates of 16 without, 7 with the median (but 68.6 % -> 63.3 % of the characters).

Photo car_18 with 5 % white and black dots. It is read correctly even without a filter: moderate noise rarely breaks the recogniser, so a filter matters only for harder photos.



none
✓ LJ67CTU



median 3x3
✓ LJ67CTU

Grainy photo (Gaussian noise, high ISO)

Filters to put into Filter 1, Filter 2 ...

```
1. Bilateral filter (edge-preserving): Spatial sigma 1.5, Range sigma 0.15
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

Grain changes every pixel a little; a median would round off thin strokes and a Gaussian blur would smear them. The bilateral filter averages only neighbours of similar brightness, so flat areas become clean and the strokes stay sharp.

What was measured

Measured: 8 -> 9 exact plates of 16 with bilateral; Gaussian 5x5 gave 5 and median 6.

Photo car_18 with Gaussian noise (sigma 0.06). Read correctly even without a filter, see above.



none
✓ LJ67CTU



bilateral
✓ LJ67CTU



Gaussian 5x5
✓ LJ67CTU



median 3x3
✓ LJ67CTU

Blurry photo (out of focus or shaky)

Filters to put into Filter 1, Filter 2 ...

```
1. Unsharp masking: Amount 1.5, Blur sigma 1.5
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

Unsharp masking makes the blurred edges steeper; Richardson-Lucy deblurring tries to reverse a Gaussian blur. Honest answer: the gain is small and not reliable, and motion blur is not repaired. Take the photo again if you can.

What was measured

Measured: 5 exact plates of 16 without; unsharp 6, Richardson-Lucy 5 (characters 64.7 % -> 62.4 % and 66.4 %).

Photo car_18 out of focus (Gaussian blur, sigma 2). Read correctly even without a filter.



none
✓ LJ67CTU



unsharp
✓ LJ67CTU



Richardson-Lucy
✓ LJ67CTU

Low contrast (haze, fog, rain, dirty glass)

Filters to put into Filter 1, Filter 2 ...

```
1. Contrast stretching: Low % 1, High % 99
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

Everything lies in a narrow band of grey levels. Stretching the 1 % to 99 % range to full black and white restores the difference between the strokes and the plate. The program does this by itself for very flat photos, so the filter is mostly for looking.

What was measured

Measured: 9 exact plates of 16 both ways; characters 81.2 % -> 83.0 %.

Photo car_18 with the contrast squeezed to 30 %. Read correctly even without a filter.



none
✓ LJ67CTU



stretching
✓ LJ67CTU

Shadow or uneven light across the photo

Filters to put into Filter 1, Filter 2 ...

```
1. Illumination correction: Blur sigma 40, Strength 1
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

A shadow across the plate hides the strokes on the dark side. Dividing by the slowly changing light brings both sides to the same level. Local histogram equalisation is the alternative. Do not add a gamma: the program already repairs the overall brightness.

What was measured

Measured: 6 exact plates of 16 without, 8 with illumination correction (characters 72.0 % -> 70.3 %). A suggested gamma made it worse (4 of 16).

Photo car_18 with the right side 4 times darker. Read correctly even without a filter.



none
✓ LJ67CTU



illumination corr.
✓ LJ67CTU

Glare or burnt-out plate (head-lamp, flash, sun)

Filters to put into Filter 1, Filter 2 ...

```
No filter: the best choice is the original photo (leave Filter 1 to Filter 4 empty).
```

Why

Pixels that are pure white carry no information any more: no filter can bring the characters back. The program refuses regions with more than a third saturated pixels (penalty). Re-take the photo from another angle or without flash.

What was measured

Not a filter problem; nothing to measure.

Photo of a screen (pixel grid, moire)

Filters to put into Filter 1, Filter 2 ...

```
1. Frequency low-pass (noise): Cut-off 0.5
```

To add them yourself: click the list of Filter 1 (it says "(none)"), choose the filter, type the numbers in brackets into the white fields under it, then do the same for Filter 2. Press "Recognise on filtered photo".

Why

A camera pointed at a monitor records the monitor's own pixel grid and colour fringes. They are regular and very fine, so a low-pass filter in the frequency domain removes them. Best of all: do not photograph the screen; open the original image file in the app.

What was measured

Not measured (no test photos of screens); a reasoned recommendation only.

Any perspective view: from the front, from the side, from above

A plate seen from the side or from above is squeezed and slanted. The program straightens it by itself (perspective correction with a homography) whenever the detected plate looks skewed; no filter can do that. To see what filters add at different views, the 16 test photos were turned with a homography about the vertical axis (yaw: the car is seen from the side) and about the horizontal axis (pitch: seen from above) and read again, with and without a filter.

Plates read exactly, of the 16 test photos turned into that view (share of correct characters in brackets):

View	no filter	CLAHE	enlarge x1.5 + unsharp
frontal (as photographed)	**8**/16 (74.6 %)	**9**/16 (70.7 %)	**9**/16 (71.5 %)
seen from the side, 30 degrees	**5**/16 (56.1 %)	**7**/16 (70.2 %)	**8**/16 (65.0 %)
seen from the side, 50 degrees	**4**/16 (48.2 %)	**3**/16 (50.6 %)	**6**/16 (58.4 %)
seen from above, 30 degrees	**5**/16 (72.7 %)	**7**/16 (66.0 %)	**4**/16 (59.8 %)

Reading the table: without a filter the plates read fall from 8 of 16 (front) to 5 at 30 degrees, 4 at 50 degrees and 5 from above. From the side, enlarging 1.5 times + mild unsharp masking gave the most plates (8 and 6); local equalisation helped at 30 degrees only. From above, local equalisation gave more plates (7) but fewer correct characters, and enlarging + sharpening did harm (4). Gains are one to three plates out of 16 and the characters sometimes went down: a hint, not a proof. The front row for the two filters comes from the earlier run on the original photos. The 6 real oblique photos of the test set read 1 of 6 without a filter, 2 of 6 with enlarging + unsharp masking and 3 of 6 with local equalisation.

frontal photo (car_18)

as photographed



none
✓ LJ67CTU



CLAHE
✓ LJ67CTU



enlarge x1.5 + unsharp
✓ LJ67CTU

from the side 30 deg



none
reads LJ67CIU



CLAHE
✓ LJ67CTU



enlarge x1.5 + unsharp
✓ LJ67CTU

from the side 50 deg



none
✓ LJ67CTU



CLAHE
reads J67C



enlarge x1.5 + unsharp
✓ LJ67CTU

from above 30 deg



none
reads LJ67CT



CLAHE
✓ LJ67CTU



enlarge x1.5 + unsharp
reads LJ67CT

oblique photo (car_30)

as photographed



none
reads HZF448



CLAHE
✓ HZF440



enlarge x1.5 + unsharp
✓ HZF440

from the side 30 deg



none
no plate



CLAHE
reads HZF448



enlarge x1.5 + unsharp
reads HZF448

from the side 50 deg



none
no plate



CLAHE
no plate



enlarge x1.5 + unsharp
no plate

from above 30 deg



none
reads HZF448



CLAHE
✓ HZF440

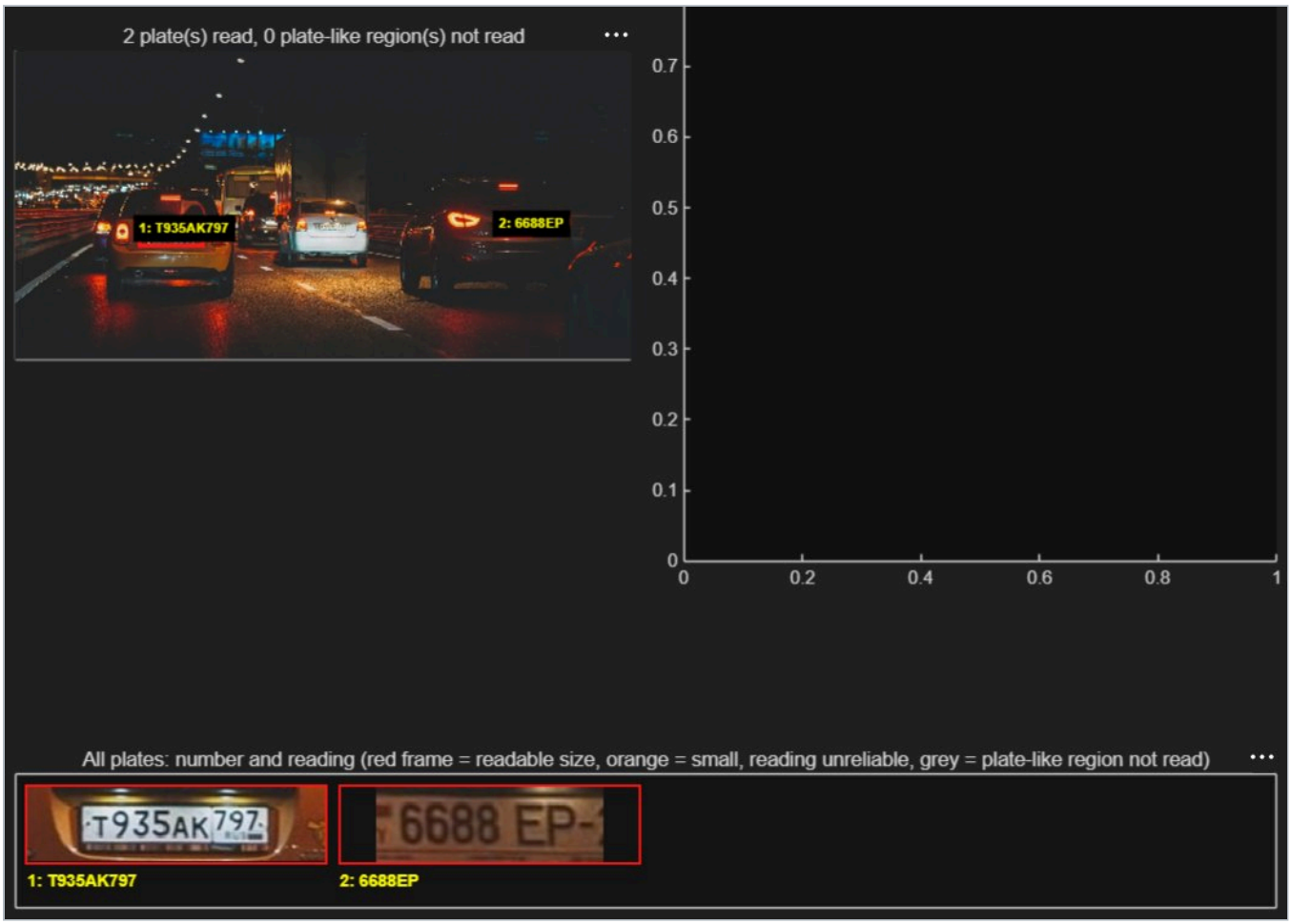


enlarge x1.5 + unsharp
✓ HZF440

What to do at any view. (1) Photograph the plate from the front whenever you can: at 50 degrees the plate is only 64 % as wide, and most readings fail whatever the filter. (2) Seen from the side (up to about 40 degrees): choose the situation "Angled plate" and press "Use recommended" (enlarge 1.5 + unsharp masking). (3) Seen from above: choose "Seen from above" (local equalisation), then press "Compare with original" and keep the filter only if the reading gets better. (4) Never use blur filters (mean, Gaussian) on an angled plate: the squeezed characters have even thinner strokes. (5) With several cars in the photo use "Pick car..." (tab 4 Plate), so the program straightens the plate of the car you want.

Many plates at once, at night and by day

Photo with several cars: choose "Many cars in the photo" or "Many cars at night" and press "Use recommended" (this ticks "Find ALL plates" in tab 4), then "Recognise on filtered photo". Every plate the program accepts gets a box in the photo (red: 90 pixels wide or more; orange: smaller, reading unreliable; grey dashed: a plate-like region that could not be read) and, under the photos, a crop with its number and reading. You can also add filters of your own first: the plates are searched in the filtered photo.



Your night traffic photo with the situation "Many cars at night": boxes in the photo, the crops with their readings under it.

Measured limits. (1) A plate needs about 100 pixels of width to be read. Plates of 20 to 30 pixels, such as the far cars of your day traffic photo (640 x 480), are neither found nor read, with or without a filter (enlarging 2 and 3 times, gamma and local equalisation were tried): only a photo with more pixels helps. (2) On your night traffic photo (1710 x 855, three plates) 2 are found: the Mini T935AK797 (right with every filter) and the dark SUV 6688 EP-2 (read 6688EP, 6688EP2 with local equalisation or enlarging 3 times, otherwise 6688F or 6688H); the white car (plate about 70 pixels) is missed. (3) On the 16 test photos a region with a plate-SVM score of at least -0.3 contained the plate in 14 of 16 photos, also when they were shrunk to 40 %, with about 1 false region per photo; the grey dashed boxes use this rule.

New in the recogniser (tested on the 16 held-out photos, exact plates before -> after: clean 8 -> 8, lit night 7 -> 8, very dark 6 -> 6, shrunk to 40 % 4 -> 4; characters 74.6 -> 74.6 %, 62.3 -> 68.6 %, 61.9 -> 61.9 %, 58.0 -> 59.0 %): (a) a dark photo is also searched in a locally equalised copy, so you do not have to choose a filter for the search; (b) when fewer than 4 characters can be cut out of a plate, characters that touch each other are cut apart at the thin necks between them, with a top-hat that removes the glare of lamps; (c) plate-like regions that could not be read are reported. These additions did no harm and gave one more plate on the lit night photos, but they did not make your dark.jpg readable: its plate (about 75 pixels, red light, lamp glare fused with the characters) is now located ("plate too small") but not read. Honest conclusion: night plates under about 100 pixels with glare are beyond this recogniser; a closer or sharper photo, or training with your own labelled night photos (markPlateBoxes, then train_real_model), is what helps.

Automatic enlargement of small plate crops (new, on by default): a plate narrower than 160 pixels is enlarged with bicubic interpolation to about 160 pixels and sharpened with an unsharp mask before its characters are cut out, which is the "Small or far plate" chain applied to the plate only. On the 16 test photos shrunk to 40 % it raised the exact plates from 4 to 7 (characters 59.0 -

> 64.1 %); on clean photos nothing changed (8 of 16, 74.6 %); on the night-lit small photos nothing changed either (5 of 16); a target of 220 pixels gave only 5, and bicubic cutting alone cost two clean plates.

Your second set of photos (Find ALL, the model the app loads; the true texts were read by eye from the photos). White BMW, Virginia plate M3PLUS1 (72 px wide): read W3PLUS1, 6 of 7 characters right (M3PLU before the automatic enlargement of small plate crops). Indian taxi, yellow plate KA53C6932 (127 px): read KA53C6332, one character wrong (9 -> 3). Scooter, two-line plate KA02K / X9181 (rows 36 px wide): only the upper row is found, and only after enlarging 2 times (read K402X or K402K for KA02K); the lower row is not found. Night street, Tesla with a California plate 1RAM11 (182 px): found only after enlarging 2 times, read 1RAI1; the other two plates (about 60 and 50 px) are not found. Night traffic: T935AK797 right, 6688EP for 6688EP2, the white car missed. Black BMW M4 (Georgia plate, about 35 px wide) and the other cars of that photo: nothing found. False reading: the sign "NO TURN ON RED" is read as TURN. Why plates are missed or misread: the characters of plates under about 100 px have only 6 to 10 pixels of height, and the character SVM was trained on 2.5 thousand characters of mostly car plates from other countries (9/3 and M/W are typical mistakes); more labelled photos of your own (markPlateBoxes, then train_real_model) are the way to improve that.

A real night photo of yours: dark.jpg

dark.jpg is a real night photo with a small plate (about 75 pixels wide) lit red by the tail lights. None of the filters makes the program read MH14 TCF 735: without a filter and with local equalisation it finds no plate; with enlarging + sharpening it finds a plate-like region but reads only "4TC5". The plate has too few pixels. The honest advice for this kind of photo is a closer or sharper photo (the plate should be at least 100 pixels wide). The pictures show the red outline of the region the program accepted, if any.



none
no plate



enlarge x2 + unsharp
reads 4TC5



CLAHE
no plate

All 17 filters, tested with their start values

Every filter was run alone, with the start values the panel put into its box at the time of the test, on the 16 clean test photos. Doing nothing reads 8 of 16 plates exactly (74.6 % of the characters). Clean photos have no problem for a filter to cure, so a filter cannot be expected to win here; the table shows which filters do harm when a beginner adds them by accident. No filter raised an error or returned values outside 0 to 1.

Filter	Exact plates	Plates found	Characters correct	Verdict
no filter	**8 / 16**	13 / 16	74.6 %	
Resize (change the resolution)	7 / 16	14 / 16	67.3 %	slightly worse
Convolution with my own kernel	7 / 16	13 / 16	69.0 %	slightly worse
Mean (box) filter	4 / 16	12 / 16	60.2 %	worse
Gaussian filter	4 / 16	11 / 16	56.0 %	worse
Median filter	7 / 16	13 / 16	69.9 %	slightly worse
Bilateral filter (edge-preserving)	6 / 16	13 / 16	67.5 %	worse
Unsharp masking	7 / 16	13 / 16	67.3 %	slightly worse
Laplacian sharpening	7 / 16	12 / 16	63.1 %	slightly worse
Deblur (Richardson-Lucy)	10 / 16	12 / 16	72.5 %	about the same
Gamma correction	7 / 16	13 / 16	68.0 %	slightly worse
Brightness / contrast	9 / 16	13 / 16	70.1 %	about the same
Contrast stretching	8 / 16	13 / 16	75.7 %	about the same
Histogram equalisation	8 / 16	13 / 16	69.6 %	slightly worse
Local histogram equalisation (CLAHE)	9 / 16	13 / 16	70.7 %	about the same
Illumination correction	8 / 16	13 / 16	69.0 %	slightly worse
Frequency low-pass (noise)	4 / 16	12 / 16	58.9 %	worse
Frequency band-pass	3 / 16	11 / 16	48.8 %	worse

Verdict: "worse" = at least 2 plates fewer than doing nothing; "slightly worse" = 1 plate fewer or at least 5 points fewer correct characters; everything else "about the same", because with 16 photos a difference of one or two plates is noise (Richardson-Lucy 10, CLAHE 9 and brightness/contrast 9 plates are inside that noise: their characters were not better). Harmful at the old start values: Gaussian 5 x 5 (4 of 16), box filter (4), frequency low-pass 0.3 (4), band-pass 0.03..0.5 (3) and bilateral (6; it is made for grainy photos, not clean ones). After this test the start values were made milder and re-tested the same way: Gaussian 3 x 3, sigma 0.8: 7 of 16 (73.7 % of the characters); low-pass 0.5: 7 of 16 (66.4 %), 0.7: 6 of 16; band-pass 0.01..0.8: 9 of 16 (71.1 %), 0.02..0.9: 6 of 16.

Reference: what every filter does

Resize (change the resolution) — Spatial resolution

How it works

Changes the number of pixels. Enlarging makes new pixels by interpolation: nearest copies the closest pixel (blocky), bilinear averages 4 neighbours (smooth), bicubic uses 16 neighbours (smoothest, sharpest edges). Shrinking first blurs a little and then drops pixels, so it removes some noise and makes everything faster.

Good for

Small or far plates: enlarge 2 times with bicubic and sharpen slightly. Measured on photos shrunk to 40 %: 4 -> 5 plates of 16 with the enlargement, 6 with enlargement + unsharp masking (+5.2 points of characters).

Be careful

It cannot create detail that the camera never recorded: a plate only 40 pixels wide stays unreadable. Do NOT shrink big photos to save time: shrinking to 0.5 cost 3 of 16 plates (8 -> 5).

Numbers: Scale 0.25 to 4 (start 1.5); Interp. 1 nearest, 2 bilinear, 3 bicubic (start 3)

Convolution with my own kernel — Spatial filtering (convolution)

How it works

A small matrix (the kernel) slides over the photo. Every output pixel is the weighted sum of the neighbours under the kernel. A kernel whose numbers add up to 1 keeps the brightness (blur, sharpen); a kernel that adds up to 0 (Laplacian, Sobel) keeps only changes, i.e. edges. The border pixels need padding (see the Border setting): replicate repeats the edge pixels, symmetric mirrors them, zero fills with black.

Good for

Learning how filters work; your own mild sharpening, for example [0 -1 0; -1 5 -1; 0 -1 0].

Be careful

Edge kernels (sum 0) turn the photo almost black: use them only to look at edges, not before recognition.

Numbers: no numbers; the matrix is typed under "Own kernel" (odd size, e.g. [0 -1 0; -1 5 -1; 0 -1 0])

Mean (box) filter — Linear smoothing filters

How it works

Replaces every pixel by the plain average of its k x k neighbours. Weak noise is averaged away, but every edge is smeared as well.

Good for

Hardly anything: very light grain only.

Be careful

Blurs every edge. Measured: even the smallest window (3 x 3) read only 4 of 16 clean test plates (60.2 % of the characters). The median or bilateral filter is better.

Numbers: Window 3 to 15 (start 3)

Gaussian filter — Linear smoothing filters

How it works

A weighted average: the centre pixel counts most, far neighbours little (bell-shaped weights, width sigma). Smoother and less blocky than the box average.

Good for

Almost nothing in this task: it is only for steps that dislike tiny noise. If you must use it, take the mild start values (3 x 3, sigma 0.8).

Be careful

Blurs the thin strokes of small characters. Measured on the clean test photos: the old start values (5 x 5, sigma 1) read 4 of 16 plates instead of 8; the new mild ones (3 x 3, sigma 0.8) read 7 of 16. On noisy photos a 5 x 5 Gaussian also cost 3 plates.

Numbers: Window 3 to 21 (start 3); Sigma 0.5 to 6 (start 0.8)

Median filter — Non-linear filters (noise)

How it works

Sorts the k x k neighbours by brightness and takes the middle value. One isolated white or black dot can never be the middle value, so it disappears; edges stay fairly sharp.

Good for

Salt-and-pepper noise: isolated white and black dots (dead sensor pixels, transmission errors, JPEG sparkle). Measured: +2 plates of 16 at 5 % dots.

Be careful

On ordinary grain it rounds off thin strokes: on Gaussian noise it cost 2 of 16 plates.

Numbers: Window 3 to 9 (start 3)

Bilateral filter (edge-preserving) — Non-linear filters (noise)

How it works

Like a Gaussian filter, but a neighbour whose brightness is very different (the other side of an edge) gets almost no weight. Flat areas are smoothed, the edges of the characters stay sharp.

Good for

Grainy photos of normal brightness (Gaussian noise). Measured: 8 -> 9 plates of 16 at sigma 0.06, while Gaussian 5 x 5 gave 5 and median 6.

Be careful

On clean photos it costs plates (6 of 16 instead of 8 at the start values): use it only when the photo is really grainy. On simulated night photos it did not help either (lit street 7 -> 6, very dark 6 -> 5; with a fixed sigma_r 4, which is why sigma_r now follows the grey range of the photo). Slow on big photos.

Numbers: Spatial sigma 0.5 to 2 (start 1.5); Range sigma 0.05 to 0.4 (start 0.15, relative to the grey range of the photo)

Unsharp masking — Sharpening and deblurring

How it works

Subtracts a blurred copy from the photo and adds the difference back: J = I + amount x (I - Gaussian(I)). Edges get steeper, so the strokes of the characters stand out.

Good for

Slightly soft photos and small plates after enlarging (amount 1, sigma 1.2).

Be careful

It also amplifies noise and draws bright and dark halos next to edges. Strong blur is not repaired by it.

Numbers: Amount 0 to 4 (start 1.5); Blur sigma 0.5 to 5 (start 1.5)

Laplacian sharpening — Sharpening and deblurring

How it works

Subtracts the Laplacian (second derivative) of the photo. It reacts to every sudden change of brightness, so fine detail and noise are both emphasised.

Good for

Quick sharpening of a clean, slightly soft photo.

Be careful

Noisier than unsharp masking: use it only on clean photos.

Numbers: Amount 0 to 3 (start 1)

Deblur (Richardson-Lucy) — Sharpening and deblurring

How it works

Richardson-Lucy deconvolution: it assumes the photo is the true picture blurred by a Gaussian of width sigma and searches, step by step (iterations), for the sharp picture that would give exactly this photo when blurred.

Good for

Out-of-focus blur of about 1 to 2 pixels (sigma 1.5, 10 iterations), in theory. In the tests it was not reliable: on clean photos 10 of 16 plates (but 72.5 % of the characters), on blurred photos 5 of 16, the same as doing nothing.

Be careful

Not for motion blur. Too many iterations amplify noise and draw ripples next to edges. Measured: no reliable gain on blurred test photos.

Numbers: Blur sigma 0.5 to 4 (start 1.5); Iterations 3 to 30 (start 10)

Gamma correction — Brightness and contrast

How it works

J = I to the power gamma. A gamma below 1 brightens the dark parts a lot and the bright parts little; above 1 darkens the photo.

Good for

To look at dark photos (gamma 0.5 to 0.7) or washed-out photos (1.3 to 1.6). On simulated night photos gamma 0.6 changed the plate count by -1 / 0 and the characters by +4.2 / -1.8 points: no clear gain, because the program repairs the brightness itself.

Be careful

The program already repairs brightness by itself; a second gamma on top over-corrects (it cost 2 of 16 plates on shaded photos).

Numbers: Gamma 0.3 to 3 (start 0.7); below 1 brightens

Brightness / contrast — Brightness and contrast

How it works

J = gain x (I - 0.5) + 0.5 + offset. A gain above 1 pushes dark and bright levels apart (more contrast); the offset moves everything brighter or darker.

Good for

Slightly flat photos.

Be careful

Bright areas can burn out to pure white (clipping) and the lost detail is gone for good.

Numbers: Contrast 0.3 to 3 (start 1.4); Brightness -0.3 to 0.3 (start 0)

Contrast stretching — Brightness and contrast

How it works

Finds the darkest 1 % and the brightest 1 % of the grey levels and stretches them to full black and full white.

Good for

Hazy, foggy or rainy photos where everything looks grey. Measured: +1.8 points of character accuracy on photos squeezed to 30 % contrast.

Be careful

It stretches the noise too, and the program already stretches the contrast by itself when it is very low.

Numbers: Low % 0 to 20 (start 1); High % 80 to 100 (start 99)

Histogram equalisation — Brightness and contrast

How it works

Re-maps the grey levels so that all levels are used about equally often (one mapping for the whole photo). A strong contrast boost.

Good for

Very flat photos, mainly to look at them.

Be careful

Exaggerates noise. One mapping for everything: a bright lamp next to a dark car gives a poor result (use local equalisation instead).

Numbers: no numbers

Local histogram equalisation (CLAHE) — Brightness and contrast

How it works

Contrast-limited adaptive histogram equalisation: the photo is cut into tiles, every tile gets its own equalising mapping (limited by the clip limit so flat areas do not explode), and neighbouring tiles are blended. Dark corners and bright spots both gain contrast.

Good for

Uneven light where a normal-brightness photo has a dark side and a bright side (typical: 8 tiles, clip limit 3). Measured on the angled plates: 3 of 6 read exactly instead of 1 of 6 (9 instead of 8 of 16 overall, but -3.9 points of characters): a hint, not a proof.

Be careful

Amplifies noise in dark flat areas: on simulated night photos it cost plates (lit street 7 -> 6, very dark 6 -> 5 and -15 points of characters). Lower the clip limit if you try it on a noisy photo.

Numbers: Tiles 2 to 16 (start 8); Clip limit 1 to 10 (start 3)

Illumination correction — Uneven illumination

How it works

Estimates the slowly changing light as a very wide Gaussian blur and divides the photo by it, so a shadow across the photo or the gradient of a lamp disappears while the letters stay.

Good for

A shadow across the plate or one side of the photo darker than the other. Measured: +2 plates of 16 (but -1.7 points of characters).

Be careful

A big dark object (the car body) is 'corrected' too, and bright sky above a dark road gets flattened. The program cannot tell a shadow from scene content, so it never suggests this filter by itself.

Numbers: Blur sigma 10 to 100 (start 40); Strength 0 to 1 (start 1)

Frequency low-pass (noise) — Frequency domain

How it works

The photo is turned into its frequencies with the Fourier transform; the fast-changing (high-frequency) parts above the cut-off are removed (Butterworth shape), then the photo is transformed back. It is a smoothing filter designed in the frequency domain.

Good for

Regular fine patterns such as the pixel grid of a screen when a monitor was photographed. Better still: open the original file instead of a photo of the screen. Not measured on real screen photos.

Be careful

Blurs the characters and draws ripples next to edges (ringing). Measured on clean test photos: cut-off 0.3 read 4 of 16 plates, 0.5 (the start value) 7 of 16, 0.7 6 of 16, against 8 without a filter.

Numbers: Cut-off 0.05 to 0.8 (start 0.5)

Frequency band-pass — Frequency domain

How it works

Keeps only a band of frequencies: the slow shading (low frequencies) and the very fine noise (high frequencies) are removed, the letter strokes lie in between and stay.

Good for

Uneven light and fine noise together, if the cut-offs are wide. With the start values 0.01 and 0.8 it read 9 of 16 clean test plates (71.1 %), about as good as doing nothing.

Be careful

A high cut-off removes the fine detail of the characters: the old start values (0.03 and 0.5) read only 3 of 16 plates (48.8 %), and 0.02 with 0.9 read 6 of 16. Keep the low cut-off at 0.01 and the high one at 0.8.

Numbers: Low cut-off 0.01 to 0.3 (start 0.01); High cut-off 0.2 to 0.9 (start 0.8)

If the plate is not read

I added filters and now nothing is read.

Look at the text under the buttons after "Recognise on filtered photo": it says whether the original photo reads. If it does, the filters made the photo worse: press "Remove filters", or choose a situation and "Use recommended". If the original does not read either, the filters are not the cause: the plate is too small (under about 90 pixels wide), too dark, burnt out or hidden. For a small plate choose "Small or far plate"; if it is small and dark, a closer photo is the only real cure.

The picture turned black, grey or flat after filtering.

Edge kernels (the presets Edge (Laplacian) and Sobel have the sum 0) and the frequency band-pass turn the photo into an edge map or a flat grey picture. The app prints "WARNING: the filtered photo is almost completely black, white or flat". Use "Sharpen" as your own kernel, or remove the filter.

A plate is boxed with 100 % confidence but the text is wrong.

The percentage is only the agreement of the character SVMs with each other; it says nothing about whether the box is a plate or the reading is right. Plates narrower than about 100 pixels are unreliable (orange box). Compare the reading with other filters and with the original.

The red box is on a van, a car body or a sign, not on the plate.

A second SVM (the character-validity SVM) now refuses regions whose pieces look like streaks and not like characters: the message is "characters look like streaks or noise". If a wrong box still appears, use "Pick car..." (tab 4 Plate) to search only inside one car. In the test, 37 of 43 non-plate regions were refused and 13 of the plates kept: it is not perfect.

My photo is .jfif, .webp, .avif or .jxl.

.jfif files and JPEGs without an extension open now. MATLAB cannot read .webp, .avif and .jxl: open them in Paint and use File > Save as > JPEG picture, or take a screenshot (Win+Shift+S) and save it as PNG.

How this was measured, and what it does not prove

Test photos: the 16 held-out photos of the project (images/real, split = test), never used for training; the numbers come from the models trained without them (lpr_svm_model_real_dev.mat). Exact plate = every character right. Simulated night: darkened photo, orange lamp colour cast, two glows and Gaussian noise 0.03 (simulateNight). Small: the photo shrunk to 40 %. Dots: salt-and-pepper 5 %. Grain: Gaussian noise sigma 0.06. Blur: Gaussian sigma 2. Low contrast: contrast squeezed to 30 %. Shadow: right side 4 times darker. Perspective: the whole photo is turned about the vertical axis (yaw) or the horizontal axis (pitch) with a homography, camera 1.6 photo widths away. These are simulations; real photographs taken at an angle differ (lens distortion, other backgrounds). The example marked "own" is your photo dark.jpg (folder Desktop\car license plate), which was not used to choose any setting. The full tables are in README.md.